

AI Medical Report Assistant

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Submitted By: Pratiksha Purkar, Shraddha Chavan

Introduction

Medical reports contain clinical terminology, abbreviations, numerical values, and reference ranges that can be difficult for patients to understand. Interpreting several test results also requires considerable time from healthcare professionals. This project develops an AI Medical Report Assistant that extracts information from uploaded reports, organizes important findings, and presents a clear, concise summary in accessible language. The system is designed to support report understanding and does not replace professional medical advice.

Problem Statement

Patients often struggle to understand abnormal values, clinical observations, and changes across reports collected over time. Existing tools may provide only a basic summary and do not combine evidence-based question answering, historical comparison, accessibility, and visualization. A secure AI solution is needed to explain reports clearly, answer questions using the uploaded documents as evidence, and help users monitor health trends without replacing professional medical advice.

Objectives

- Build a Retrieval-Augmented Generation (RAG) assistant that explains reports, summarizes key findings, and answers user questions with report-based context.
- Extract content from digital and scanned reports using document parsing and Optical Character Recognition (OCR).
- Identify abnormal values and compare current results with previous reports to reveal changes and trends.
- Present results through a health dashboard and timeline visualization.
- Support doctor-question suggestions, voice explanations, multiple languages, and report export.

Methodology

The system accepts medical reports in PDF or image format. OCR and document parsers extract text, tables, test values, units, reference ranges, and observations. Cleaned content is divided into chunks, converted into embeddings, and stored in a vector database. When a user asks a question, the RAG pipeline retrieves relevant report passages and supplies them to an LLM to generate a grounded explanation or summary. Rule-based analysis flags out-of-range values, while historical records are aligned by test name and date for comparison, dashboard charts, and timeline views. Additional modules generate suggested questions for a doctor, translate responses, produce voice output, and export a structured report. Access controls, secure storage, disclaimers, and response validation support responsible use.

Project Phases

Phase 1	Requirement Analysis	Phase 2	System Design
Phase 3	Frontend Development	Phase 4	Backend Development
Phase 5	OCR Integration	Phase 6	AI & NLP Processing
Phase 7	Dashboard & Visualization	Phase 8	Testing
Phase 9	Deployment	Phase 10	Documentation & Future Enhancements

Technologies Used

Python, Pandas, OCR and PDF-processing tools, NLP, LLMs, RAG, semantic embeddings, a vector database, speech and translation services, visualization libraries, a web framework, and a secure database.

Expected Outcomes and Applications

The system will provide grounded explanations, concise summaries, interactive question answering, abnormal-value highlighting, and comparisons across historical reports. Its dashboard, timeline, multilingual and voice support, doctor-question suggestions, and export facility will make health information more accessible and organized for patients and healthcare professionals.

Conclusion

The AI Medical Report Assistant combines OCR, structured analysis, RAG, historical comparison, and accessible interaction features in one platform. It helps users understand and track their reports while preserving privacy, grounding answers in source documents, and encouraging consultation with qualified healthcare professionals.